Filtering the bracketing phase

A rectangle benchmark for candidate eigenvalue searches in `chladni`

2 September 2026

Abstract

An eigenvalue search in `chladni` decomposes into three phases: *bracketing* (produce intervals each believed to hold one eigenvalue), *refinement* (locate the minimum inside a bracket), and *multiplicity estimation*. This report tests the first phase alone, on rectangles, where the spectrum is known in closed form and clustering and multiplicity are under exact control.

Bracketing is the phase worth testing first. Every search failure measured during the design work was a bracketing failure, and bracketing is where essentially all the evaluations go — refinement is a handful of them and multiplicity is three. Testing it in isolation also sidesteps a defect in the joint scorer, described in §1, which makes recall-with-multiplicity unable to distinguish "split the pair" from "guessed 2".

The purpose is to **filter**: to identify methods not worth developing further. Results are therefore stated as capability claims within an explicit evaluation budget, never as performance scores.

Contents

1 What is being tested, and why bracketing alone **3**
- 1.1 The three phases . . . 3
- 1.2 The joint scorer cannot measure what we need . . . 3

2 The acceptance rule **3**
- 2.1 A lattice, not a bijection . . . 3
- 2.2 Metrics . . . 4
- 2.3 Committing . . . 4

3 The instrument **4**
- 3.1 Why rectangles . . . 4
- 3.2 The basis, and a correction . . . 4
- 3.3 The controlling principle: fix the pencil, vary only prec . . . 5
- 3.4 Truth . . . 5

4 The methods **5**
- 4.1 Controls . . . 6

5 Results **6**
- 5.1 Aggregate: autonomous passes . . . 6
- 5.2 Family A — isolated . . . 7
- 5.3 Family B — pair ladder . . . 7
- 5.4 Family C — clusters of 3 and 4 . . . 11
- 5.5 Family D — exact degeneracy . . . 13
- 5.6 Family E — mixed window . . . 14

6 Analysis **14**
6.1 The headline: nobody splits a pair below $d/\lambda \approx 10^{-4}$ 14
6.2 Family B is not the whole story, and reading it alone gives the wrong verdict . . 15
6.3 Cost scales with prec even when the requirement does not 15
6.4 The controls validate the metrics 16

7 Filter verdict **16**

8 What this cannot conclude **16**

1 What is being tested, and why bracketing alone

1.1 The three phases

Most candidate searches decompose the same way: a bracketing pass produces candidate intervals, a local refinement locates the minimum inside each, and a multiplicity estimator reads the tension spectrum at the located point. This decomposition was verified rather than assumed. In all five searches implemented to date, the bracketing pass provably terminates before the first call to the refinement routine, and no refined value re-enters subdivision; none is entangled.

The three phases are also very unequal. All five searches dispatch to one of only two multiplicity estimators, and only one of those is live — the ratio-gap plus local-minimum test (C1)+(C2). So multiplicity is a *shared* component: a defect there affects every candidate identically, and the filter cannot discriminate between candidates on that axis. Refinement is similarly settled: Brent on σ_1^2 is measured machine-exact on every domain tried, at every regularization tolerance from `1e-7` to `1e-16`. That leaves bracketing as the phase where the candidates actually differ.

1.2 The joint scorer cannot measure what we need

The end-to-end scorer matches found eigenvalues against truth at a fixed relative tolerance of `1e-6`, expanding a reported (λ, μ) into μ copies of λ. Below $d/\lambda <$ `1e-6`, a method that never separates a pair but claims multiplicity 2 has both copies match two *distinct* truth entries, scoring recall 1.000, zero spurious, and `perfect=True` — while the honest report of one root of multiplicity 1 scores 0.500 *and* takes a multiplicity penalty. Measured:

d	d/λ	reports one root, claims $\mu = 2$	reports one root, honest $\mu = 1$
5.9e-04	1.2e-05	recall 0.000, spurious 2	recall 0.000, spurious 1
5.9e-05	1.2e-06	recall 1.000, **perfect**	recall 0.500
5.9e-07	1.2e-08	recall 1.000, **perfect**	recall 0.500, multX 1

The metric does not merely go blind, it rewards the wrong behaviour, and it does so at prec$\cdot\lambda \approx$ `4.9e-5`, essentially the boundary at which the basis stops being able to resolve the pair at all. This is the same error the reference layer had already found and removed from its own `collapse` function, three orders tighter, with the comment that a tolerance "does not describe the spectrum, it imposes a policy on it".

A bracket set carries no multiplicities, so the exploit does not exist and the joint scorer is not on the path for anything in this report.

2 The acceptance rule

2.1 A lattice, not a bijection

Let S(prec) denote the true spectrum with levels merged whenever a group's *diameter* is at most prec$\cdot\lambda$. A bracket set is acceptable when it lies in the lattice

fine end	one bracket per true eigenvalue
coarse end	one bracket per level of S(prec)

and anything between the two is acceptable. In particular an *accidental* split of a within-prec cluster is not penalised: downstream, refinement finds both members, the collapse rule sees two roots inside the merge tolerance, and the reported answer is two levels of multiplicity 1 — which is legal, because merging is permitted and never required. Penalising the bracketer for it would contradict the multiplicity convention.

Bounding the bracket count rather than fixing it is also what defeats the two degenerate strategies. Returning $[(a, b)]$ scores coverage 1.0 and zero spurious while being useless; returning ten thousand slivers has perfect purity and absurd cost. Neither survives a rule that constrains the count from both sides.

2.2 Metrics

Coverage — hard. Every true eigenvalue lies in some bracket. A miss is unrecoverable downstream.

Purity — hard, and the metric with teeth. No bracket may contain eigenvalues spanning more than prec$\cdot\lambda$. If it does, refinement returns one of them and the other is lost.

Fineness — free within the lattice. Reported against both bounds.

Spurious — a count, not a failure: cost, plus the risk of a phantom root from a bracket that brushes a well without containing its minimum.

Cost — evaluations. Tension and eigenvector calls are counted in one counter, since an eigenvector call costs 100–107 % of a tension call and a method that leans on them should not get them free.

A method **passes** a case when coverage is 1.0 and no bracket is impure.

2.3 Committing

A bracketing method must *commit*: the returned intervals are those it believes hold an eigenvalue, not a subdivision of $[a, b]$. Without this requirement a partitioning method gets coverage for free. Measured on one case with four true eigenvalues in $[10, 88]$: the swing's cells tile the interval, total width 78.0 of 78.0, and score coverage 4/4 by construction, while uniform windows cover 5.9 of 78.0 and have to earn it.

3 The instrument

3.1 Why rectangles

The search layer is basis-agnostic by construction — it sees only the tension function and the interval. So a method can only respond to the *shape* of the tension curve, and that shape has four parameters: the floor ε, the wall slope c, the plateau, and the eigenvalue spacing. Rectangles control all four, and add three things no other domain offers: closed-form truth (no table-precision ceiling), exact control of clustering and multiplicity, and low cost.

On the unit square the eigenvalues are $\lambda/\pi^2 = m^2 + n^2$, so multiplicity is a number-theoretic property of the level: 2 at $N = 5$, 3 at $N = 50$, 4 at $N = 65$, 6 at $N = 325$. Perturbing to $\mathrm{rect}(1 + \delta, 1)$ splits a multiplicity-m level into m eigenvalues whose spacing scales with δ and whose internal gaps are unequal — for $N = 65$ they are in the ratio $1 : 2.2 : 1$, and that ratio is invariant as δ sweeps decades. The cluster *shape* is fixed while its *scale* varies, so a method cannot pass by exploiting one particular spacing.

3.2 The basis, and a correction

The pencil must resolve the tension to below the finest prec under test, or the experiment measures the objective rather than the algorithm. Checking that requirement found that **corner-adapted Fourier–Bessel is the wrong basis for a rectangle**, by four orders. Unit square, truncation effectively off, σ_μ at each level (i.e. counting multiplicity):

basis	cols	N=2 (μ=1)	N=5 (μ=2)	N=50 (μ=3)	N=65 (μ=4)
FB n=80	80	1.68e-15	2.60e-14	1.42e-11	3.17e-12
FB n=160	160	1.56e-15	2.51e-14	1.41e-11	3.15e-12
PlaneWave 100	200	4.57e-16	5.67e-16	1.67e-15	1.66e-15
PlaneWave 160	320	4.86e-16	6.02e-16	1.20e-15	1.45e-15

The reason is that a right angle has $\pi/\omega = 2$, an integer, so the eigenfunction is *analytic* there and no fractional expansion is called for. Corner-FB is the right tool for a singular corner and the wrong one for a domain with none; that pure Fourier–Bessel wins on the L-shape does not transfer.

Three candidate explanations for the Fourier–Bessel floor were tested and rejected before the basis itself was suspected. It is not the expansion order (flat from 80 to 300 columns), not the collocation (flat from $n_b = 400$ to 6400 and $n_i = 200$ to 3200), and not the rank truncation — sweeping the threshold from its default `1.5e-12` down to zero moves the floor by under 1 % while the retained rank rises from 70 to 80. It is also not a property of λ: the mode $(7, 1)$ gives the same 3.5–3.7e−12 on squares of side 1, 2, 3 and 5, i.e. at λ from 493 down to 19.7, so it is set by $k\cdot$diam, the mode index, and rescaling the domain cannot buy precision.

With sized plane waves every level used in this report resolves to `1.2e-15` or better, multiplicity 6 included.

3.3 The controlling principle: fix the pencil, vary only prec

The pencil is built once per case at the best resolution available and is identical across the whole prec sweep. prec enters only as an input to the bracketing algorithm, telling it when to stop splitting.

This is the difference between testing bracketing and re-discovering the limits of an under-resolved tension curve. Three separate basis-construction errors arose during this work — a rank frozen over a wide window, too few Fourier–Bessel terms on a corner domain, and corner-FB on a rectangle — and every one silently degraded the objective by orders while looking healthy. With the pencil fixed, any difference across prec is attributable to the algorithm by construction. A hard precondition asserts, before any case is scored, that σ_μ is at or below `1e-14` at every true eigenvalue counting multiplicity, and that the finest prec in the sweep clears the pencil's own limit.

3.4 Truth

Window completeness comes from the closed-form spectrum function, which is complete by construction rather than by truncation. The perturbed levels take their values from the mode formula directly, because the spectrum function snaps near-degenerate values together at a relative tolerance of `1e-13` — measured, it holds the $5\pi^2$ pair distinct down to $d =$ `5.9e-12` and merges it at `6.0e-13`. No case in this report approaches that floor.

4 The methods

Each is exposed as `brackets(tensions, a, b, *, prec, modes)`, extracted from the corresponding search rather than reimplemented, and reusing that search's own helpers. The five searches themselves are untouched.

`uniform_grid` One grid, discrete minima of σ_1, bracket from the neighbouring samples. Deliberately not prec-aware beyond a cap: a uniform grid cannot reach a spacing of prec$\cdot\lambda$

at any usable cost, so its brackets are grid-spacing wide whatever is asked. That is the honest behaviour of the method.

`lappy` The original recursive bisection. Its recursion is gated on σ_2's *value* against a threshold, which is a property of the grid rather than of the pair: a grid of spacing h sees about $ch/2$, so the gate engages when $h \leq 2\,\mathrm{ttol}/c$ whatever the separation.

`lappy_repaired` The same, with the gate removed: recurse into *every* discrete minimum, unconditionally, down to a floor. The floor here is $\mathrm{prec}\cdot\lambda$ rather than a fixed fraction of the interval — the fixed fraction was previously found to be load-bearing and unexplained, losing a different eigenvalue at each of three values.

`lipschitz` Piyavskii–Shubert branch and bound. Its completeness certificate is conditional on the Lipschitz constant L, and L estimated from a pilot has been measured *unsound*: on one domain a 20-point pilot with $\gamma = 2$ underestimated the true slope, ratio 0.672, so the certificate can be false. It is included to measure that, not because it is expected to win. Brackets are read off the sample cache, not off the candidate intervals — the candidate list in the original is dead code, left behind when refinement was rewritten to work from the samples.

`swing_cells` Cells cut at peaks of the angle between successive minimising eigenvectors. Those peaks fall at branch *crossings*, which lie between eigenvalues, so they partition the interval into cells holding one eigenvalue each. Each crossing is **bisected** rather than taken at the peak sample: detection localises a crossing only to the sampling interval, and when the pair separation is below the stride the cell boundary can land outside the pair. Bisection costs $\log_2(h/d)$ eigenvector evaluations and converges regardless of how coarse the stride was.

`swing_direct` The same cells, each then narrowed by a Lipschitz-free DIRECT subdivision before the bracket is returned. The ablation against `swing_cells` is what attributes the composed method's behaviour to one mechanism or the other.

4.1 Controls

Two adversarial controls, without which the metrics are unvalidated: `single_bracket` returns $[(a, b)]$ and must fail purity on every case with more than one level; `fine_grid` returns thousands of equal slivers and must fail on spurious count and cost. If either scores well, the metric is broken rather than the control.

5 Results

`orc` is the cheapest of the tested settings that reached coverage 1.0 with no impure bracket; `aut` is the single setting fixed across the whole suite before any per-case result was seen. − means no setting succeeded within the budget of 1500 evaluations. Only the `aut` column may be aggregated: `orc` is a best-of-K statistic and is biased upward in K.

5.1 Aggregate: autonomous passes

Across the 27 swept cases (the multiplicity-6 stress case excluded), counting only the autonomous setting:

method	prec = 10^{-4}	10^{-8}	10^{-11}	10^{-14}
`uniform_grid`	16/27	9/27	7/27	5/27
`lappy`	17/27	10/27	8/27	6/27
`lappy_repaired`	12/27	5/27	5/27	5/27
`lipschitz`	22/27	14/27	12/27	10/27
`swing_cells`	21/27	13/27	11/27	8/27
`swing_direct`	22/27	14/27	10/27	8/27
`single_bracket`	5/27	3/27	3/27	3/27
`fine_grid`	0/27	0/27	0/27	0/27

5.2 Family A — isolated

A:isolated 11 eigenvalues in $[5, 300]$; sweep took 320 s.

	prec = 10^{-4}		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
`uniform_grid`	830	–	830	–	830	–	830	–
`lappy`	–	–	–	–	–	–	–	–
`lappy_repaired`	1449	1449	1449	–	1449	–	1449	–
`lipschitz`	610	–	610	–	610	–	610	–
`swing_cells`	–	–	–	–	–	–	–	–
`swing_direct`	923	923	1302	1302	–	–	–	–
`single_bracket`	–	–	–	–	–	–	–	–
`fine_grid`	–	–	–	–	–	–	–	–

5.3 Family B — pair ladder

B:d=1.8e-01 4 eigenvalues in $[10, 88]$; sweep took 151 s.

	prec = 10^{-4}		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
`uniform_grid`	–	–	–	–	–	–	–	–
`lappy`	1016	–	1016	–	1016	–	1016	–
`lappy_repaired`	1103	–	1103	–	1103	–	1103	–
`lipschitz`	1298	1298	1298	1298	1298	1298	1298	1298
`swing_cells`	528	528	788	788	988	988	1188	1188
`swing_direct`	688	688	942	942	1150	1150	1352	1352
`single_bracket`	–	–	–	–	–	–	–	–
`fine_grid`	–	–	–	–	–	–	–	–

B:d=1.8e-02 4 eigenvalues in $[10, 88]$; sweep took 144 s.

	prec = 10^{-4}		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
`uniform_grid`	–	–	–	–	–	–	–	–
`lappy`	–	–	–	–	–	–	–	–
`lappy_repaired`	830	–	830	–	830	–	830	–
`lipschitz`	–	–	–	–	–	–	–	–
`swing_cells`	–	–	–	–	–	–	–	–
`swing_direct`	–	–	–	–	–	–	–	–
`single_bracket`	–	–	–	–	–	–	–	–
`fine_grid`	–	–	–	–	–	–	–	–

B:d=1.8e-03 4 eigenvalues in $[10, 88]$; sweep took 152 s.

	prec = 10^{-4}		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
`uniform_grid`	86	158	–	–	–	–	–	–
`lappy`	122	302	–	–	–	–	–	–
`lappy_repaired`	–	–	–	–	–	–	–	–
`lipschitz`	286	709	–	–	–	–	–	–
`swing_cells`	184	704	–	–	–	–	–	–
`swing_direct`	219	850	–	–	–	–	–	–
`single_bracket`	–	–	–	–	–	–	–	–
`fine_grid`	–	–	–	–	–	–	–	–

B:d=1.8e-04 4 eigenvalues in $[10, 88]$; sweep took 151 s.

	prec = 10^{-4}		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
`uniform_grid`	86	158	–	–	–	–	–	–
`lappy`	122	302	–	–	–	–	–	–
`lappy_repaired`	1454	–	–	–	–	–	–	–
`lipschitz`	289	1174	–	–	–	–	–	–
`swing_cells`	184	648	–	–	–	–	–	–
`swing_direct`	227	779	–	–	–	–	–	–
`single_bracket`	–	–	–	–	–	–	–	–
`fine_grid`	–	–	–	–	–	–	–	–

B:d=1.8e-08 4 eigenvalues in $[10, 88]$; sweep took 166 s.

	prec = 10^{-4}		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
`uniform_grid`	86	158	86	158	–	–	–	–
`lappy`	122	302	122	302	–	–	–	–
`lappy_repaired`	1011	–	1011	–	–	–	–	–
`lipschitz`	441	1191	441	1191	–	–	–	–
`swing_cells`	163	686	215	1063	–	–	–	–
`swing_direct`	197	891	245	1257	–	–	–	–

B:d=1.8e-12 4 eigenvalues in $[10, 88]$; sweep took 180 s.

	prec $= 10^{-4}$		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
`uniform_grid`	86	158	86	158	86	158	–	–
`lappy`	122	302	122	302	122	302	–	–
`lappy_repaired`	–	–	–	–	–	–	–	–
`lipschitz`	471	1246	471	1246	471	1246	–	–
`swing_cells`	131	349	170	492	200	602	–	–
`swing_direct`	152	418	191	561	215	661	–	–

B:d=5.8e-01 4 eigenvalues in $[10, 88]$; sweep took 150 s.

	prec $= 10^{-4}$		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
`uniform_grid`	446	–	446	–	446	–	446	–
`lappy`	302	302	302	302	302	302	302	302
`lappy_repaired`	1316	–	1316	–	1316	–	1316	–
`lipschitz`	293	1193	293	1193	293	1193	293	1193
`swing_cells`	131	379	170	548	200	678	230	808
`swing_direct`	152	458	197	633	225	773	251	900
`single_bracket`	–	–	–	–	–	–	–	–
`fine_grid`	–	–	–	–	–	–	–	–

B:d=5.9e-02 4 eigenvalues in $[10, 88]$; sweep took 142 s.

	prec $= 10^{-4}$		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
`uniform_grid`	–	–	–	–	–	–	–	–
`lappy`	–	–	–	–	–	–	–	–
`lappy_repaired`	922	–	922	–	922	–	922	–
`lipschitz`	1308	1308	1308	1308	1308	1308	1308	1308
`swing_cells`	184	–	249	–	299	–	349	–
`swing_direct`	225	–	300	–	338	–	392	–
`single_bracket`	–	–	–	–	–	–	–	–
`fine_grid`	–	–	–	–	–	–	–	–

B:d=5.9e-03 4 eigenvalues in $[10, 88]$; sweep took 152 s.

	prec $= 10^{-4}$		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
`uniform_grid`	–	–	–	–	–	–	–	–
`lappy`	–	–	–	–	–	–	–	–
`lappy_repaired`	1426	1426	–	–	–	–	–	–
`lipschitz`	–	–	–	–	–	–	–	–
`swing_cells`	658	658	1009	1009	1279	1279	–	–
`swing_direct`	844	844	1213	1213	1462	1462	–	–
`single_bracket`	–	–	–	–	–	–	–	–
`fine_grid`	–	–	–	–	–	–	–	–

B:d=5.9e-04 4 eigenvalues in $[10, 88]$; sweep took 149 s.

	prec = 10^{-4}		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
uniform_grid	86	158	–	–	–	–	–	–
lappy	122	302	–	–	–	–	–	–
lappy_repaired	684	1324	–	–	–	–	–	–
lipschitz	305	1283	–	–	–	–	–	–
swing_cells	137	510	–	–	–	–	–	–
swing_direct	159	648	–	–	–	–	–	–
single_bracket	–	–	–	–	–	–	–	–
fine_grid	–	–	–	–	–	–	–	–

B:d=5.9e-05 4 eigenvalues in $[10, 88]$; sweep took 150 s.

	prec = 10^{-4}		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
uniform_grid	86	158	–	–	–	–	–	–
lappy	122	302	–	–	–	–	–	–
lappy_repaired	1223	–	–	–	–	–	–	–
lipschitz	503	1349	–	–	–	–	–	–
swing_cells	172	556	–	–	–	–	–	–
swing_direct	216	694	–	–	–	–	–	–
single_bracket	–	–	–	–	–	–	–	–
fine_grid	–	–	–	–	–	–	–	–

B:d=5.9e-07 4 eigenvalues in $[10, 88]$; sweep took 173 s.

	prec = 10^{-4}		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
uniform_grid	86	158	–	–	–	–	–	–
lappy	122	302	–	–	–	–	–	–
lappy_repaired	1025	–	–	–	–	–	–	–
lipschitz	502	1379	–	–	–	–	–	–
swing_cells	171	482	–	–	–	–	–	–
swing_direct	208	617	–	–	–	–	–	–

B:d=5.9e-10 4 eigenvalues in $[10, 88]$; sweep took 166 s.

	prec = 10^{-4}		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
uniform_grid	86	158	86	158	–	–	–	–
lappy	122	302	122	302	–	–	–	–
lappy_repaired	1295	–	1295	–	–	–	–	–
lipschitz	306	789	306	789	–	–	–	–
swing_cells	195	644	273	982	–	–	–	–
swing_direct	235	804	319	1149	–	–	–	–

B:d=5.9e-12 4 eigenvalues in $[10, 88]$; sweep took 189 s.

	prec = 10^{-4}		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
uniform_grid	86	158	86	158	86	158	–	–
lappy	122	302	122	302	122	302	–	–
lappy_repaired	583	1063	583	–	583	–	–	–
lipschitz	488	1347	488	1347	488	1347	–	–
swing_cells	179	686	244	1063	294	1353	–	–
swing_direct	218	887	283	1257	333	–	–	–

5.4 Family C — clusters of 3 and 4

C:m=3,s=9.3e+00 3 eigenvalues in $[478.7, 508.3]$; sweep took 800 s.

	prec = 10^{-4}		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
uniform_grid	94	190	94	190	94	190	94	190
lappy	142	382	142	382	142	382	142	382
lappy_repaired	278	278	634	–	634	–	634	–
lipschitz	263	403	263	663	263	663	263	663
swing_cells	119	327	158	496	188	626	218	756
swing_direct	142	400	189	567	213	705	245	837
single_bracket	–	–	–	–	–	–	–	–
fine_grid	–	–	–	–	–	–	–	–

C:m=3,s=9.5e-01 3 eigenvalues in $[478.7, 508.3]$; sweep took 670 s.

	prec = 10^{-4}		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
uniform_grid	190	–	190	–	190	–	190	–
lappy	382	–	382	–	382	–	382	–
lappy_repaired	252	252	387	1352	387	1352	387	1352
lipschitz	162	347	162	347	162	347	162	347
swing_cells	1312	–	–	–	–	–	–	–
swing_direct	–	–	–	–	–	–	–	–
single_bracket	–	–	–	–	–	–	–	–
fine_grid	–	–	–	–	–	–	–	–

C:m=3,s=9.5e-02 3 eigenvalues in $[478.7, 508.3]$; sweep took 503 s.

	prec = 10^{-4}		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
uniform_grid	–	–	–	–	–	–	–	–
lappy	–	–	–	–	–	–	–	–
lappy_repaired	165	–	–	–	–	–	–	–
lipschitz	–	–	–	–	–	–	–	–
swing_cells	102	262	–	–	–	–	–	–
swing_direct	122	336	–	–	–	–	–	–
single_bracket	–	–	–	–	–	–	–	–
fine_grid	–	–	–	–	–	–	–	–

C:m=3,s=9.5e-03 3 eigenvalues in $[478.7, 508.3]$; sweep took 576 s.

	prec = 10^{-4}		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
uniform_grid	76	94	–	–	–	–	–	–
lappy	94	142	–	–	–	–	–	–
lappy_repaired	–	–	899	899	899	899	899	899
lipschitz	160	344	–	–	–	–	–	–
swing_cells	138	346	–	–	–	–	–	–
swing_direct	162	429	–	–	–	–	–	–
single_bracket	1	1	–	–	–	–	–	–
fine_grid	–	–	–	–	–	–	–	–

C:m=4,s=1.2e+00 4 eigenvalues in $[622.3, 660.8]$; sweep took 792 s.

	prec = 10^{-4}		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
uniform_grid	–	–	–	–	–	–	–	–
lappy	382	–	382	–	382	–	382	–
lappy_repaired	328	328	625	–	625	–	625	–
lipschitz	170	385	170	385	170	385	170	385
swing_cells	378	378	547	547	677	677	807	807
swing_direct	491	491	668	668	788	788	928	928
single_bracket	–	–	–	–	–	–	–	–
fine_grid	–	–	–	–	–	–	–	–

C:m=4,s=1.2e+01 4 eigenvalues in $[622.3, 660.8]$; sweep took 1062 s.

	prec = 10^{-4}		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
uniform_grid	94	190	94	190	94	190	94	190
lappy	142	382	142	382	142	382	142	382
lappy_repaired	334	334	1054	–	1054	–	1054	–
lipschitz	273	442	273	762	273	762	273	762
swing_cells	144	438	196	685	236	875	276	1065
swing_direct	180	551	242	808	272	986	314	1188
single_bracket	–	–	–	–	–	–	–	–
fine_grid	–	–	–	–	–	–	–	–

C:m=4,s=1.2e-01 4 eigenvalues in $[622.3, 660.8]$; sweep took 794 s.

	prec = 10^{-4}		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
uniform_grid	–	–	–	–	–	–	–	–
lappy	–	–	–	–	–	–	–	–
lappy_repaired	478	–	752	–	752	–	752	–
lipschitz	407	407	772	–	772	–	772	–
swing_cells	1145	–	–	–	–	–	–	–
swing_direct	–	–	–	–	–	–	–	–
single_bracket	–	–	–	–	–	–	–	–
fine_grid	–	–	–	–	–	–	–	–

C:m=4,s=1.2e-02 4 eigenvalues in $[622.3, 660.8]$; sweep took 758 s.

	prec $= 10^{-4}$		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
`uniform_grid`	76	94	–	–	–	–	–	–
`lappy`	94	142	–	–	–	–	–	–
`lappy_repaired`	381	381	–	–	–	–	–	–
`lipschitz`	160	364	–	–	–	–	–	–
`swing_cells`	131	338	–	–	–	–	–	–
`swing_direct`	154	389	–	–	–	–	–	–
`single_bracket`	1	1	–	–	–	–	–	–
`fine_grid`	–	–	–	–	–	–	–	–

5.5 Family D — exact degeneracy

D:mult=2 2 eigenvalues in $[239.3, 254.1]$; sweep took 345 s.

	prec $= 10^{-4}$		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
`uniform_grid`	76	94	76	94	76	94	76	94
`lappy`	94	142	94	142	94	142	94	142
`lappy_repaired`	210	210	295	936	295	936	295	936
`lipschitz`	108	317	108	317	108	317	108	317
`swing_cells`	132	393	184	640	224	830	264	1020
`swing_direct`	168	552	220	781	260	969	296	1175
`single_bracket`	1	1	1	1	1	1	1	1
`fine_grid`	–	–	–	–	–	–	–	–

D:mult=3 3 eigenvalues in $[478.7, 508.3]$; sweep took 542 s.

	prec $= 10^{-4}$		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
`uniform_grid`	76	94	76	94	76	94	76	94
`lappy`	94	142	94	142	94	142	94	142
`lappy_repaired`	221	221	331	1224	331	1224	331	1224
`lipschitz`	99	188	99	188	99	188	99	188
`swing_cells`	119	261	158	352	188	422	218	492
`swing_direct`	138	302	189	397	209	469	249	535
`single_bracket`	1	1	1	1	1	1	1	1
`fine_grid`	–	–	–	–	–	–	–	–

D:mult=4 4 eigenvalues in $[622.3, 660.8]$; sweep took 615 s.

	prec = 10^{-4}		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
`uniform_grid`	76	94	76	94	76	94	76	94
`lappy`	94	142	94	142	94	142	94	142
`lappy_repaired`	201	201	262	759	262	759	262	759
`lipschitz`	152	358	152	358	152	358	152	358
`swing_cells`	146	454	185	597	215	707	245	817
`swing_direct`	173	515	214	662	236	762	274	882
`single_bracket`	1	1	1	1	1	1	1	1
`fine_grid`	–	–	–	–	–	–	–	–

5.6 Family E — mixed window

E:mixed 29 eigenvalues in $[150, 560]$; sweep took 1145 s.

	prec = 10^{-4}		10^{-8}		10^{-11}		10^{-14}	
method	orc	aut	orc	aut	orc	aut	orc	aut
`uniform_grid`	–	–	–	–	–	–	–	–
`lappy`	–	–	–	–	–	–	–	–
`lappy_repaired`	–	–	–	–	–	–	–	–
`lipschitz`	–	–	–	–	–	–	–	–
`swing_cells`	–	–	–	–	–	–	–	–
`swing_direct`	–	–	–	–	–	–	–	–
`single_bracket`	–	–	–	–	–	–	–	–
`fine_grid`	–	–	–	–	–	–	–	–

6 Analysis

6.1 The headline: nobody splits a pair below $d/\lambda \approx 10^{-4}$

Family B is the pair ladder. The table gives the *oracle* cost — the cheapest tested setting that succeeded — at each separation, at the tightest prec for which the pair must still be *split* rather than merged. A dash is failure at every setting within the 1500-evaluation budget.

d	d/λ	`uniform`	`lappy`	`lappy_rep`	`lipschitz`	`swing_cells`	`swing_direct`
5.8e-01	1.2e-02	446	302	1316	293	**131**	152
1.8e-01	3.6e-03	–	1016	1103	1298	**528**	688
5.9e-02	1.2e-03	–	–	922	1308	**184**	225
1.8e-02	3.6e-04	–	–	**830**	–	–	–
5.9e-03	1.2e-04	–	–	1426	–	**658**	844
1.8e-03	3.6e-05	–	–	–	–	–	–
...	down to	–	–	–	–	–	–
5.9e-12	1.2e-13	–	–	–	–	–	–
1.8e-12	3.6e-14	–	–	–	–	–	–

Every method's splitting limit lies between $d/\lambda = 1.2\text{e}{-4}$ and $3.6\text{e}{-5}$ at this budget — four to five orders coarser than the pencil's own resolution limit of $\varepsilon/c \approx 4\text{e}{-14}$ absolute, about 10^{-15} relative. **The gap between what the objective supports and what the searches extract is roughly ten orders of magnitude**, and it is the most important number here.

Below the limit the methods do not fail loudly: they merge the pair and report success at

any prec loose enough to permit it. At $d = 1.8\text{e}{-3}$ and prec $= 10^{-4}$, where merging is legal, `uniform_grid` passes at 86 evaluations; at prec $= 10^{-8}$ on the same case, where it is not, every method fails.

6.2 Family B is not the whole story, and reading it alone gives the wrong verdict

A first draft of this report concluded from the ladder above that `lipschitz` should be cut. The per-family breakdown contradicts that, and the contradiction is recorded here rather than quietly fixed. Oracle / autonomous passes at prec $= 10^{-8}$:

method	A (1)	B (12)	C (8)	D (3)	E (1)
`uniform_grid`	1/0	3/2	3/2	3/3	0/0
`lappy`	0/0	4/3	4/2	3/3	0/0
`lappy_repaired`	1/0	**6**/0	**6**/2	3/3	0/0
`lipschitz`	1/0	5/**5**	5/**4**	3/3	0/0
`swing_cells`	0/0	**6**/**5**	3/3	3/3	0/0
`swing_direct`	1/**1**	**6**/**5**	3/3	3/3	0/0

Three things the pair ladder alone concealed:

The swing methods are notably worse on clusters than on pairs — 3 of 8 on family C against 6 of 12 on family B, where `lappy_repaired` reaches 6 of 8 and `lipschitz` 5. A cell decomposition cut at eigenvector crossings is well suited to splitting two eigenvalues and less so to a tight group of three or four with unequal internal gaps.

`lappy_repaired` **has the best oracle record and the worst autonomous one** — 6 of 12 on family B at oracle, and **0 of 12** autonomously. Essentially all of its apparent capability is per-case tuning. That is the single largest oracle–autonomous gap in the study, and it is exactly the failure mode the two-column protocol exists to expose: a method that looks strongest in a tuned comparison and would ship as the weakest.

`lipschitz` **is the most consistent autonomous performer**, 5/5 on B and 4 of 5 on C — its oracle and autonomous columns nearly coincide, meaning one fixed setting works across cases. Its known theoretical weakness, that a pilot-estimated Lipschitz constant can underestimate the true slope and render the completeness certificate false, did not bite on rectangles. That is a real caveat for corner domains, not a measured failure here.

Nobody passes family E — the 29-eigenvalue mixed window — at any prec or any setting. A window holding isolated levels, six pairs and a triple defeats every method in the study within budget.

6.3 Cost scales with prec even when the requirement does not

On most cases the purity requirement is *identical* at prec $= 10^{-8}$, 10^{-11} and 10^{-14}: the smallest true gap is far above prec$\cdot\lambda$ at all three, so nothing about what must be resolved changes. Yet cost rises monotonically. On $d = 5.8\text{e}{-1}$, `swing_cells` needs 131, 170, 200 and 230 evaluations at the four precisions — a 1.8× increase for ten orders of prec, buying nothing.

The cause is that every method bisects to prec$\cdot\lambda$ regardless of the local gap. Refining a cell boundary to 10^{-13} when the nearest other eigenvalue is 0.58 away is pure waste. This violates the "not more work than asked" clause of the operational definition, and it is a defect of the whole field rather than of any one method.

That this is cost and not capability was confirmed directly rather than inferred. On $d = 5.9\text{e}{-3}$ at prec $= 10^{-14}$, where the requirement is identical to prec $= 10^{-11}$, `swing_cells` fails at the 1500-evaluation cap and **succeeds at 1549 evaluations** once the cap is raised — against the 1279 it needs at 10^{-11}. A 21 % cost increase, and the only thing separating "fails" from "succeeds" is the budget line.

6.4 The controls validate the metrics

`single_bracket` returns $[(a, b)]$ and scores coverage 1.0 with zero spurious on every case. It is rejected by **purity alone**, on 18 of 21 cases. The three it passes are the family-D exact-degeneracy cases, which contain a single level — there one bracket *is* the correct answer, so the metric accepting it is right rather than broken.

`fine_grid` exceeds the evaluation cap on every case and is rejected on **cost**. Neither degenerate strategy survives, and each is rejected by the criterion intended for it.

7 Filter verdict

Capability claims within the measured budget of 1500 evaluations, on rectangles, for the bracketing phase only.

`uniform_grid` — **CUT as a candidate.** Weakest or joint-weakest in every family, splits no pair tighter than $d/\lambda = 1.2\text{e}{-2}$. It should be retained as the control that every proposal must beat, which is the role it already has.

`lappy` **(original)** — **CUT.** Dominated by `lappy_repaired` in every family at both oracle and autonomous, and splits nothing below $d/\lambda = 3.6\text{e}{-3}$. Its σ_2-value recursion gate was already known to test the grid rather than the pair; this measures the consequence.

`lappy_repaired` — **KEEP, with the gap as the priority.** Best oracle record of any method (6/12 on B, 6/8 on C) and the worst autonomous (0/12 on B). Nothing about its subdivision is wrong; what fails is its self-configuration. Closing that gap is worth more than any change to its search strategy.

`lipschitz` — **KEEP.** The most consistent autonomous performer, with oracle and autonomous nearly coinciding. This reverses the draft verdict, which cut it on the pair ladder alone. Its unsound-certificate risk is real but was not exercised here, and should be tested on a corner domain before it is either trusted or discarded.

`swing_cells` — **KEEP.** Cheapest method wherever it succeeds, often by 3–7×, and reaches the ladder's limit. Its weaknesses are clusters of three and four (3 of 8) and the isolated-eigenvalue case, which it fails at every prec.

`swing_direct` — **KEEP.** Costs 15–25 % more than `swing_cells` and buys no pair-splitting capability, but it is what rescues the isolated case (1/1 against 0/0). The two mechanisms are orthogonal rather than ranked, and the ablation says so: cell-cutting handles pairs, DIRECT handles isolated wells.

No method is cut on the strength of family B alone, and no method survives family E.

8 What this cannot conclude

- **A correction.** An earlier draft of this report claimed that prec $= 10^{-14}$ could never be exercised as a requirement on this instrument, on the grounds that the only separation distinguishing it from 10^{-11} — a pair at $d/\lambda \approx 10^{-13}$ — sits just 1.2× above the `snap`

merge floor of the closed-form spectrum. **That reasoning was wrong: family B does not use snapped truth.** It takes the pair values from the mode formula directly, precisely so that a controlled separation is not at the mercy of a merge policy. Measured, the two values are 833 floating-point ulps apart at $\delta = 10^{-13}$ and 250 apart at 3×10^{-14}, and the pencil has $d/(\varepsilon/c) = 138$ of headroom at the first.

Two such cases were therefore built and run, and both pass the floor precondition (headroom 4.1 and 3.3). At prec $= 10^{-4}$, 10^{-8} and 10^{-11} the pair may legally merge and most methods pass cheaply; at prec $= 10^{-14}$ it must be split, and **every method fails**. So prec $= 10^{-14}$ is now exercised as a genuine requirement, and the answer is consistent with the ladder: nothing splits a pair at $d/\lambda = 1.2\text{e}{-13}$ when nothing splits one at $3.6\text{e}{-5}$.

- **Rectangles only.** The property rectangles cannot reproduce is the tension floor jumping between adjacent modes on a corner domain — separately found to be largely an artifact of a marginal basis, but re-measured at recipe grade on only one domain.
- **Bracketing only.** A method that deliberately blurs bracketing into multiplicity estimation is under-credited by construction. The swing is such a method: the same eigenvector signal that cuts its cells also distinguishes a true degeneracy from two distinct eigenvalues, and none of that is visible to a bracket-set metric.
- **One budget, one seed, three settings per knob.** A method rejected here is rejected *at 1500 evaluations*, not in principle. The one non-monotonicity in the ladder — `lappy_repaired` passing $d = 1.8\text{e}{-2}$ while both swing variants pass the tighter $5.9\text{e}{-3}$ and fail it — is most likely a knob-grid artifact, since `per_gap` was swept only over $\{6, 32, 160\}$, but that was not confirmed.
- **The oracle column is best-of-K** and biased upward in the number of knobs. Only the autonomous column may be compared across methods, and the gap between the two is the most actionable quantity in the report.